GroundStateDuo – Final Report

1. Team and plan

We computed the H_2 energy curve with exact diagonalisation, VQE and DFT. All three give a bond length within about 2% of each other, differ in binding energy by up to about 1 eV (mostly because the Qiskit basis set is tiny), and disagree most when the bond is stretched, where default (spin-restricted) DFT fails.

Item	Details
Team name	GroundStateDuo
Member 1	Thomas Adesoye Adeoye
Member 2	Damilare Jeremiah Ojo
Project title	Potential Energy Curve of H_2 with DFT and a Quantum Algorithm
Forked repository	github.com/tadeoye137/qff26xUIQC

Objective. Compute the ground-state energy of H_2 against the H–H distance with exact diagonalisation (the reference), VQE (Qiskit, noise-free simulator) and DFT (Quantum ESPRESSO); extract the bond length, vibration frequency and binding energy from each curve; and compare them with each other and with experiment.

Motivation. A University of Illinois course has students compute the energy curve of two protons with DFT, molecular dynamics and a quantum computer [1]. H_2 is small and well known (bond length 0.741 Å), so every result can be checked, and stretched H_2 is where DFT is known to struggle.

Plan and delivery. Done: exact and VQE curves (23 distances, 0.5 to 3.0 Å); the DFT scan with spin-restricted and spin-broken-symmetry settings; the comparison figure and table. The isolated H-atom DFT reference was attempted but was numerically unstable, so it was not used (sections 2.3 and 4). The Morse-fit with molecular dynamics and noisy-VQE stretch goals in our submitted plan were not attempted because time did not permit.

2. Methodology

All three methods solve the same problem: the lowest electronic energy of two protons held a distance R apart (the Born–Oppenheimer approximation). Repeating this for many R gives the potential energy curve E(R).

2.1 Exact diagonalisation and VQE (Qiskit). We used PySCF to build the H_2 Hamiltonian in the minimal STO-3G basis (2 spatial orbitals, 4 spin-orbitals), mapped it to qubits with the Parity mapper and 2-qubit reduction, and added the nuclear repulsion. The exact reference came from NumPyMinimumEigensolver, which diagonalises the 4×4 qubit Hamiltonian. VQE used a UCCSD ansatz started from the Hartree–Fock state, with the SLSQP optimiser (at most 200 iterations), zero initial parameters and a noise-free StatevectorEstimator. We ran 23 distances from 0.5 to 3.0 Å, and a fine grid of 16 points from 0.60 to 0.90 Å (step 0.02 Å) for the curvature.

2.2 DFT (Quantum ESPRESSO). Quantum ESPRESSO 7.4.1, PBE functional, ultrasoft pseudopotential H.pbe-rrkjus_psl.1.0.0.UPF, plane-wave cutoffs 60 Ry (wavefunction) and 480 Ry (density), a 24 bohr (12.7 Å) cubic box, Γ point, fixed occupations, SCF convergence 10^{-8} Ry, and 23 distances from 0.5 to 3.0 Å (step 0.02 Å from 0.60 to 0.90). Two settings were run at every distance: *spin-restricted* (nspin = 1), where both electrons share one spatial orbital, and *spin-broken-symmetry* (nspin = 2, zero total magnetisation), where the two H atoms start with opposite spin so the electrons can separate when the bond stretches. The plane-wave cutoff was checked with a convergence test. Runs used 48 cores on the ICTP-EAIFR cluster in Rwanda. The second team member independently ran the same DFT scripts on a local computer and obtained the same results.

2.3 Quantities extracted. Near the minimum we fitted a quartic polynomial to E(R) (16 fine-grid points) and read off three numbers:

- **Bond length R_e:** the R where the fitted curve is lowest.
- **Vibration frequency:** the force constant is the curvature of the fit, and the wavenumber follows from the reduced mass μ = m(H)/2, half the mass of one H atom:

Force constant $k = d^2E/dR^2$ at $R = R_e$, and vibration wavenumber = $\sqrt{(k/\mu)} / (2\pi c)$, where c is the speed of light.

- **Binding energy D_e:** the energy of the separated atoms minus the energy at the minimum. For the Qiskit curves the atom energy is E(H) = −0.466582 Ha (STO-3G, PySCF UHF), so $D_e = 2E(H) - E(\text{min})$. For DFT, an isolated spin-polarised H atom was

numerically unstable with this pseudopotential (energies between −0.4999 and −0.5161 Ha depending on settings), so D_e is taken as E(R = 3 to 4 Å, spin-broken) − E(min), where the molecule has effectively dissociated.

Conversions: 1 Ha = 27.211386 eV = 2 Ry. All code (h2_analysis.py, h2_dft.sh and job scripts) and the CSV data are in the forked repository.

3. Results

3.1 VQE reproduces the exact answer. At every distance the VQE energy lay within 5×10^{-7} Ha of exact diagonalisation (differences from 10^{-11} to 4.2×10^{-7} Ha), so for this small problem the quantum algorithm is as good as the reference. Exact and VQE are therefore one curve in the figure and one row in the table.

3.2 Three methods compared. Figure 1 shifts each curve so its own minimum is zero, so only the shape is compared.

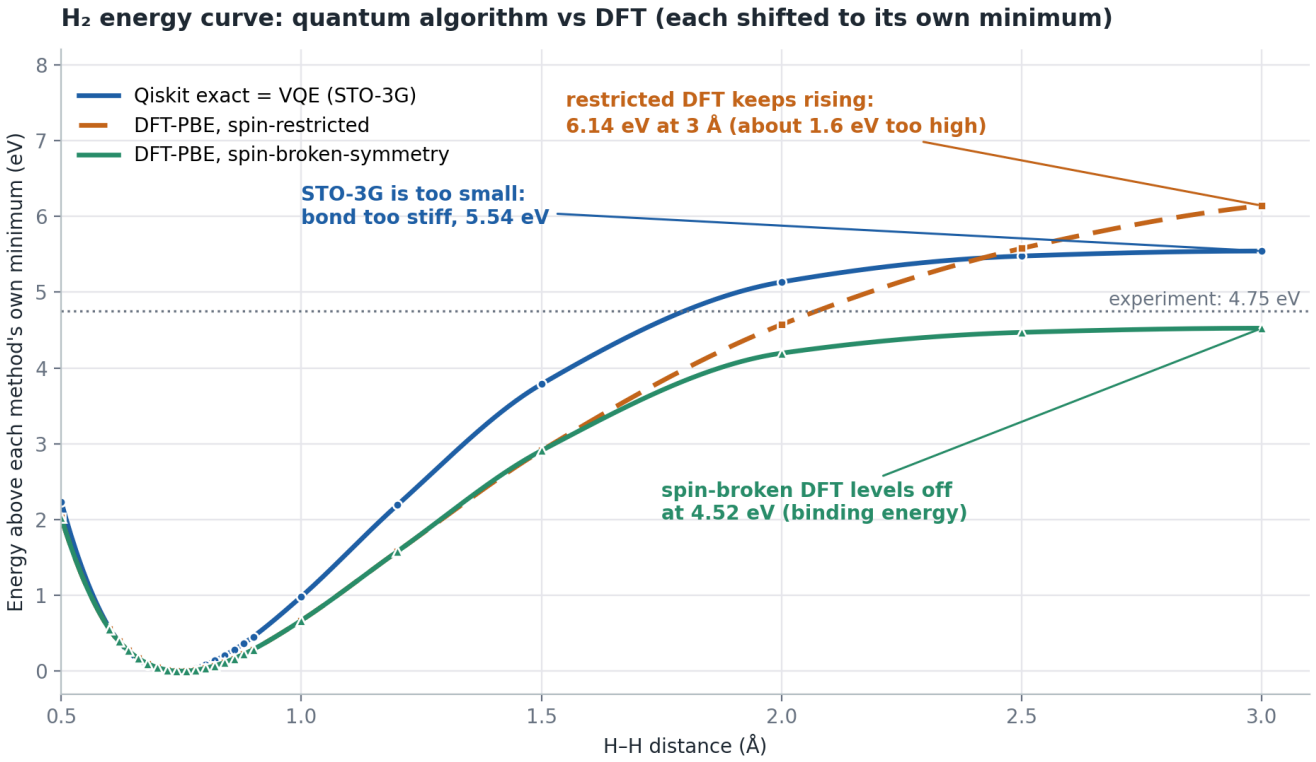


Figure 1. Energy relative to each curve's minimum (eV) against H–H distance. Qiskit exact and VQE lie on top of each other. The dashed line marks the experimental binding energy, 4.75 eV.

Near the minimum the three curves are close. At large R they separate: the spin-restricted DFT curve keeps climbing instead of levelling off, while the Qiskit and spin-broken DFT curves flatten.

3.3 Numbers from the fits.

Read this table with care. The Qiskit rows use the small STO-3G basis and the DFT row uses a much larger plane-wave basis, so differences between them mix basis-set error with method error. Each row can fairly be compared with experiment, but the table is not a ranking of VQE against DFT.

Method	Bond length R_e (Å)	Force constant k (N/m)	Vibration (cm^{-1})	Binding energy D_e (eV)
Exact diagonalisation and VQE (identical; STO-3G)	0.7346	746.7	5015	5.555
DFT, PBE, spin-broken	0.750	about 553	4315	about 4.5 (4.52 at 3 and 4 Å)
Experiment	0.7414	575	4401	4.747

The DFT force constant is computed from the quoted wavenumber with μ = m(H)/2. At the equilibrium bond the restricted and spin-broken DFT energies are almost the same (−1.166541 and −1.166544 Ha at 0.74 Å), so the bond length and vibration above apply to both.

3.4 The stretched bond. At 3.0 Å the spin-restricted DFT energy is −0.940908 Ha, but the spin-broken energy is −1.000341 Ha, a drop of 0.0594 Ha (1.6 eV). Measured from each method's minimum, restricted PBE puts the dissociated atoms 6.14 eV above the molecule and spin-broken PBE 4.52 eV above. The exact STO-3G curve at 3.0 Å is already within 0.0005 Ha of two separate atoms (−0.933632 against 2 × −0.466582 = −0.933164 Ha). The 5 and 6 Å spin-broken points were discarded because the energy kept falling, which is not physical.

4. Discussion

Where the two kinds of error come from. The Qiskit numbers are exact for the model we gave them, so their gap to experiment is entirely the basis set. STO-3G uses only two orbitals, which makes the bond too stiff and too strong: bond length 0.9% short, vibration

14% high, binding energy 17% high (5.555 against 4.747 eV). VQE does not add to this error, since it matches exact diagonalisation to 5×10^{-7} Ha. DFT has a much larger basis, so its error is a different kind: the approximate functional. Near the minimum it is close to experiment (bond length 1.2% long, vibration 2% low, binding energy about 5% low), which is why the two methods agree where the bond is normal.

Why restricted DFT fails when the bond is stretched. A spin-restricted calculation forces both electrons into the same spatial orbital. That is right near equilibrium, but when the atoms move apart the correct picture is one electron on each atom. The restricted calculation instead keeps a share of the "both electrons on one atom" configuration, which costs real energy and is the reason the curve rises instead of flattening (static correlation and delocalisation error in the language of the field). The result is a dissociation energy too large by about 1.6 eV at 3 Å.

What spin-broken DFT does and does not fix. Allowing the two atoms to carry opposite spin lets the electrons separate, and the curve flattens with a binding energy of about 4.5 eV, within 0.25 eV of experiment. The price is that the state is not a pure singlet: it is a mixture of singlet and triplet, so its energy is not a clean quantum-mechanical eigenvalue. It is a good fix for the energy, not a correct wavefunction. The Qiskit approach has no such problem, because it treats both electrons together in a many-body wavefunction.

Limits of this study.

- The Qiskit runs are noise-free simulations. A real device adds noise, and we did not test that.
- The isolated H atom in DFT was unstable with this pseudopotential, so the DFT binding energy rests on the stretched molecule (3 and 4 Å, both 4.52 eV) rather than on an atom calculation. The value is consistent with the binding energy typically reported for PBE, which supports it.
- The vibration frequency is harmonic (from the curvature only), with no anharmonic correction.

Take-away. For H_2 the quantum algorithm and exact diagonalisation are indistinguishable, and DFT is accurate near equilibrium but needs broken spin symmetry to describe a stretched bond. A natural next step is the same comparison with a larger basis (for example 6-31G), simulated noise and a run on IBM hardware.

5. Assistance Declaration

- **Claude (Anthropic).** Learning and coding assistant: explained concepts (binding energy, spin-broken DFT, basis sets), helped write and debug the scripts and figures, and helped write this report. We ran and checked all calculations ourselves and can explain the methods, code and results.
- **Software.** Qiskit, Qiskit Nature, Qiskit Algorithms, PySCF, Quantum ESPRESSO and the PSlibrary pseudopotential, used as described in section 2 and cited in the references; no code was copied beyond library examples and documentation. The UIQC Fall Fest starter materials were also consulted.
- **ICTP-EAIFR cluster (Rwanda).** Compute time for the DFT calculations (48 cores); Dr. Akin-Ojo gave us access.
- **Mentorship.** Sophie Engineer (IBM Quantum Developer Advocate) advised by email on choosing a project; this shaped our choice of VQE on H_2. She did not review our code or this report.
- **Team contribution.** Thomas Adesoye Adeoye: Qiskit and DFT calculations and analysis, figures and report. Damilare Jeremiah Ojo: independently ran the same DFT scripts on a local computer and reproduced the results.

6. References

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